

Human Health, Sustainability

Why Bioplatforms Matter, and Can Even Save the World

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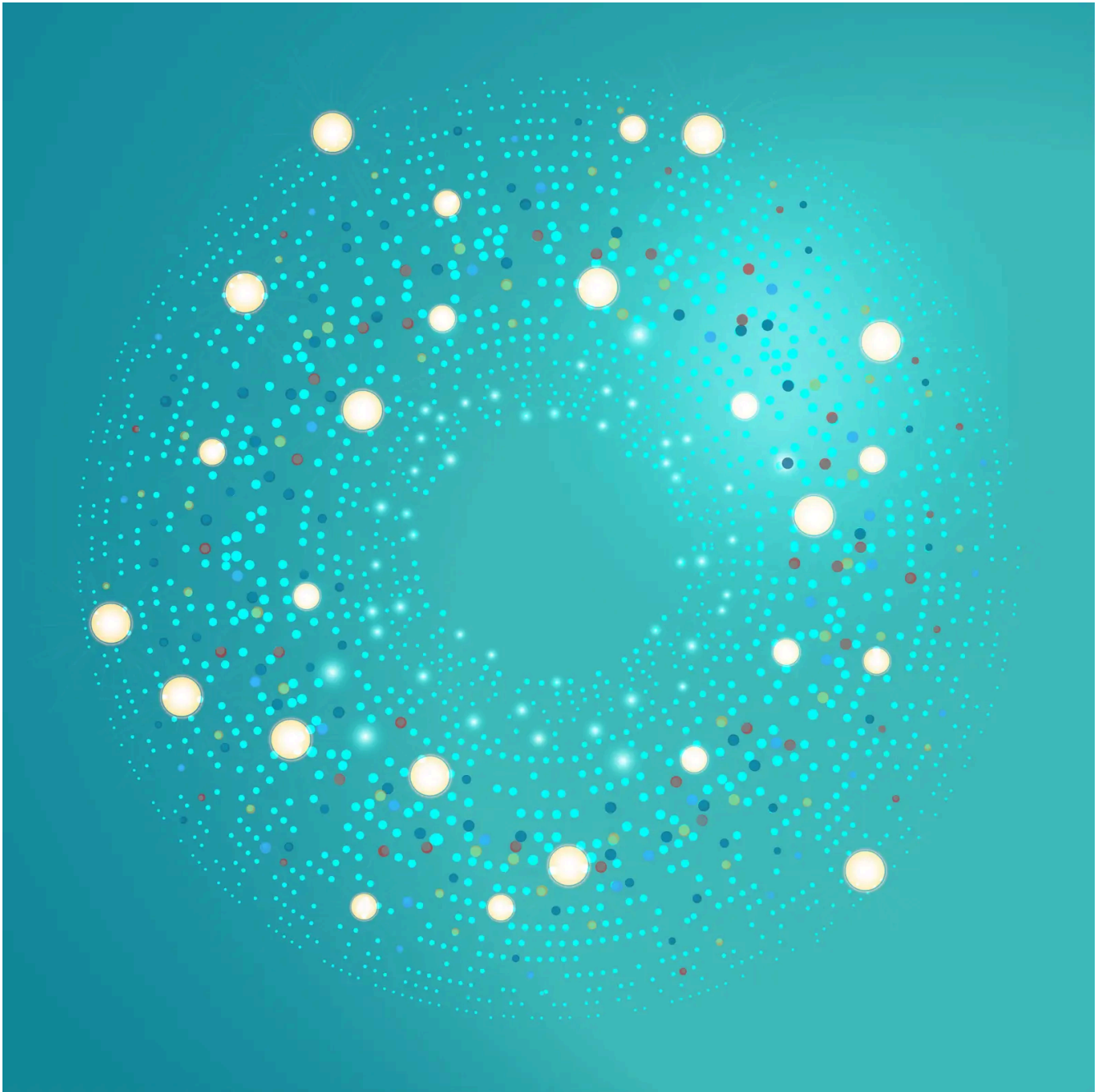


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The COVID-19 pandemic powerfully demonstrated that human ingenuity and biotechnology could be harnessed to respond to and contain a global, emergent, biological threat in an unbelievably short amount of time (less than a year).

But this wasn't simply a matter of luck, or spontaneous invention and brute force execution. This ability to understand, devise, manufacture, test and distribute antidotes and vaccines to a novel virus was due in large part to the development and maturation of bioplatforms.

What are bioplatforms and why do they matter? In the world of tech, platforms are relatively well-understood as a core framework and ecosystem in which multiple, and even limitless applications can be created. Think of Apple's iOS, a single operating system on which a virtually unlimited number of apps can be built, or Salesforce.com, a single cloud-based architecture which can serve countless business functions and industries.

In the biotechnology world, platforms are understandably less well defined and understood. At Flagship, we define bioplatforms as biotechnologies that -- once created and harnessed -- allow for the intentional and repeatable generation of multiple medicines or agricultural and sustainability products. Two of the core tenets of bioplatforms are that they: a) must serve a diversity and breadth of applications; and b) they must be universal. The more a bioplatform satisfies these tenets, the more powerful and valuable it can become. The bioplatform becomes the proverbial goose that can predictably, and with some degree of ease and constancy, lay golden egg after golden egg.

It is also important to note what a bioplatform is *not*. Although the instantiation and deployment of a bioplatform will often require the assemblage and interconnection of multiple technologies and disciplines, that assemblage alone does not create a bioplatform. Also, the consolidation of multiple drug programs, projects and products under one roof does not constitute a bioplatform, although again, the exploitation of a successful bioplatform will result in the parallel pursuit of multiple product programs. If the aforementioned tenets aren't met, and all that exists are a suite of tools or a "bag of products", a bioplatform doesn't really exist and by extension, neither do the efficiencies and exponential value that a bioplatform can unleash.

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Take for instance Moderna, and its messenger RNA (mRNA) therapeutic bioplatfrom. It was devised, designed and developed to create an mRNA molecule to encode virtually any therapeutic protein, for use as a medicine or a vaccine. The company had developed the core, universal (meaning application-agnostic) capability to design, make and test mRNAs, and has been applying it to the development of multiple vaccines (including against cytomegalovirus, Zika, respiratory syncytial virus, influenza, and others), antiviral antibodies, cancer drugs, and rare disease enzymes. Already, the bioplatfrom's value becomes imminently apparent in the sheer number and diversity of programs that a relatively small company like Moderna can pursue in parallel. Even more striking and consequential was the company's ability to rapidly direct the bioplatfrom to the creation of novel proteins, such as the SARS-CoV-2 spike protein, within just two days of the protein's sequence being released. The company and platfrom did not have to be drastically modified to tackle the pandemic, beyond having to dramatically scale up to meet the manufacturing and fill-finish demands of supplying countries around the globe with hundreds of millions of doses, exemplifying the true power of a genuinely universal platfrom.

Compare this to large pharmaceutical organizations that aren't based on bioplatfroms, but pride themselves on being vertically integrated companies spanning biological discovery to commercialization. Few if any were able to respond as quickly as Moderna to the pandemic, and those that did actually partnered with bioplatfrom companies to instantiate their vaccines or therapies.

An additional and important benefit of bioplatfroms is the ability to categorize their application space into sub-modalities (categories of use cases), whereby success with a couple of examples in that category dramatically and predictably unlocks the likelihood of success with other products in that category. This is why the scientists at Moderna and the National Institutes of Allergy and Infectious Disease(NIAID) were confident that an mRNA-based vaccine to a newly discovered viral protein such as the SARS-CoV-2 spike protein would very

likely work; prior preclinical and clinical successes with Moderna's Zika and cytomegalovirus mRNA vaccines, indicated that they could rely on animal models and virus neutralization experiments to predict the efficacy and safety of any new mRNA vaccine they designed.

This ability to gain confidence in large application categories is not as easily afforded by small molecule or traditional protein-based therapeutics. In those drug classes, success with a small molecule against a given target indeed tells you that drugging that target is possible, but doesn't dramatically improve the likelihood of success or speed when tackling the next new target (only slightly if the target is of the same ilk, such as with kinase targets, for example). Bioplatforms thereby render discontinuous problem landscapes into more continuous and easily navigable terrains, facilitating lateral moves once a beachhead has been established in a given application area.

Interestingly and excitingly, our continued understanding and mastery of biology is yielding a plethora of new bioplatforms, including nucleic acid-based, cell-based, and computationally-designed modalities that can produce smarter, more rationally-designed medicines, more quickly. Where the dual bioplatfrom tenets of application breadth and universality afford us limitless possibilities to tackle virtually any biological problem. This is propelling us into a golden age of biology and medicine, in which we are less and less reliant on serendipity for finding cures, and are now capable of engineering and generating creative solutions to help us all live longer, better and more productive lives.

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